

Bridging Molecular Simulations and AlphaFold to Explore Conformational Changes in Proteins

AlphaFold (AF) enables highly accurate protein structure prediction but often fails to capture the conformational changes involved in protein function. To address this limitation, we developed an enhanced sampling scheme that integrates the diffusion-based generative model used in AlphaFold3 (AF3) with concepts from molecular dynamics (MD) simulations. By introducing a repulsive biasing potential between predicted structures, the developed scheme enables AF3 to sample multiple conformational states, including ligand-bound conformational states that are rarely captured with the default AF3 settings. We also demonstrate that diverse conformations generated by AF can serve as starting points for rare-event MD sampling, enabling further investigation of protein conformational dynamics and thermodynamics.